# Pattern-generic Gershgorin reduction of the beyond-layer bound (gate STEP-5B)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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## 1. Purpose and scope

Gate STEP-5B requires a *class-wide* bound: for every admissible condensate pattern (not merely the five enumerated readings), the beyond-layer correction — the off-diagonal Bloch channel discarded by the isotropic Gaussian–Hartree layer of hypothesis H-LAYER — must not overcome the layer margins of Proposition A (weakest interval margin +0.00432, the band figure of Math437 v1.2 after the Math440 precision fix). This note (i) proves two pattern-independent lemmas (Gershgorin–Schur row bound; second-order log-det envelope with an isotropic trace identity), (ii) reduces STEP-5B to the remaining weighted additive-energy gap G1‴-AE on an extreme corner of pattern space (the spectral-control gap G1″-M4 is CLOSED BY STRUCTURE by the $P^2$-representation theorem below; G2 bookkeeping is CLOSED at second-cumulant level), and (iii) certifies all entering constants numerically at the production anchor, reproducing the Math434-audit calibration. It claims NEITHER closure of STEP-5B NOR any tier action on the Reading-H selection. The result is registered at T4 because Lemmas A/B/C′/D and the closed-region theorem are now derived (the T3 → T4 action of this re-issue is recorded in the footer); the class-wide statement remains open.

## 2. Setting

Work at the corrected-convention production anchor ($\mu^2 = 0.005$, $r_{\mathrm{braz}} = K(q_0) = \mu^2$; claim A1-KERNEL-CONV), with the dressed isotropic diagonal

$$D(k) \;=\; \hat{r} + C\,(k^2 - q_0^2)^2, \qquad \hat{r} \;=\; r_R + 2\lambda' I, \qquad \lambda' \;=\; 3u + 30vM,$$

where $r_R = 0.3045257$, $M = M(\hat{r})$ self-consistently ($M_R = 0.109414$ at $I = 0$), $u = -0.86$, $v = +3.24$, $C = 1$, $q_0 = 0.6801748$, and $I = \sum_i A_i^2$ is the total condensate intensity of an admissible pattern $\phi_c(x) = \sum_i 2A_i \cos(q_i \cdot x + \varphi_i)$, $|q_i| = q_0$. Note $\lambda' = 8.0551 > 0$ at the anchor: the pattern Hartree term *raises* the diagonal. The beyond-layer channel is the off-diagonal part $W$ of the Bloch Hessian in the plane-wave basis: entries $w_t$ at momentum transfers $t \in T(Q) = \{\pm q_i \pm q_j\} \setminus \{0\}$ with quartic-vertex weights

$$w_t \;=\; \lambda' \sum_{(i,j):\, \pm q_i \pm q_j = t} A_i A_j$$

(sextic $5vA^4$-class transfers enter at $O(w_4)$; gap G2). Define $X = D^{-1/2} W D^{-1/2}$ and the weighted transfer norm $W_{\ell^1}(P) = \sum_{t \neq 0} |w_t|$.

## 3. Two pattern-independent lemmas

**Lemma 1** (Gershgorin–Schur row bound). *For the symmetric kernel $X$ above, with $D_{\min} = \hat{r}$,*

$$\|X\| \;\le\; \sup_k \sum_{t\neq 0} \frac{|w_t|}{\sqrt{D(k)\,D(k+t)}} \;\le\; W_{\ell^1}(P)\, g(\hat{r}), \qquad g(\hat{r}) \;=\; \frac{1}{\hat{r}} \;+\; \frac{1}{\sqrt{\hat{r}\,(\hat{r}+64Cq_0^4)}}.$$

*Proof.* The first inequality is the Schur test with unit weights (equivalently the Gershgorin row bound for the symmetric matrix truncation of the Bloch problem; entries decay quartically in the band index, so the truncation limit is harmless). For the second: each transfer term is maximised when both legs are on-shell, giving $1/\hat{r}$; retaining additionally the degenerate-endpoint term $1/\sqrt{\hat{r}(\hat{r}+64Cq_0^4)}$ of the Math434 LAM geometry makes $g$ an upper envelope for every $|t| \le 2q_0$ uniformly in the row, since any further leg sits at distance $\ge 64Cq_0^4$ above the gap in the quartic well. Summing over transfers yields the claim. □

**Lemma 2** (Second-order log-det envelope; isotropic trace identity). *If $\|X\| \le a < 1$, the beyond-layer free-energy correction per unit volume obeys*

$$\left|\delta F_{\rm off}\right| = \tfrac{1}{2V}\left|\mathrm{tr}\big(\ln(\mathbf{1}+X) - X\big)\right| \;\le\; \frac{\mathrm{tr}\,X^2/V}{4\,(1-a)}, \qquad \frac{\mathrm{tr}\,X^2}{V} = \sum_{t\neq 0} |w_t|^2\, J(|t|),$$

$$J(|t|) \;=\; \int \frac{d^3k}{(2\pi)^3}\, \frac{1}{D(k)\,D(k+t)},$$

*which depends on the pattern only through the moduli $|t|$ (isotropy of $D$).*

*Proof.* Eigenvalues $x_i$ of $X$ lie in $[-a, a]$; the scalar Taylor remainder gives $|\ln(1+x) - x| \le x^2/(2(1-a))$ on that interval; summing over the spectrum and dividing by $2V$ gives the first display (the $\frac{1}{2}$ from the Gaussian free energy). The trace identity is direct: $\mathrm{tr}\,X^2 = \sum_k \sum_t w_t^2/(D(k)D(k+t)) \to V \sum_t w_t^2 J(|t|)$, and $J$ depends only on $|t|$ because $D$ is a function of $|k|$ alone. The first off-diagonal order $\mathrm{tr}\,X$ vanishes (no $t = 0$ entry in $W$). □

**Lemma 3** (Transfer matching — exact $\ell^1$ identity and caps). *For any admissible single-shell pattern ($Q = \{q_1, \ldots, q_n\}$ distinct, amplitudes $A_i > 0$, $I = \sum_i A_i^2$, $S = \sum_i A_i$):*

- *(i) for every fixed transfer $t \neq 0$ and fixed sign pair, the pairs $(i, j)$ with $\pm q_i \pm q_j = t$ form a partial matching ($q_j$ is uniquely determined by $q_i$ and $t$); hence the multiplicity obeys $m_t \le 2n$ and, by Cauchy–Schwarz along the matching, $|w_t| \le 2\lambda' I$;*
- *(ii) $\sum_{t\neq 0} |w_t| \;\le\; \lambda'\,(4S^2 - 2I) \;\le\; \lambda'\,(4nI - 2I)$, with equality in the first inequality iff no two ordered sign-pairs produce coinciding transfers (verified to machine precision on planar rings $n = 8..32$).*

*Proof.* (i) Uniqueness of $q_j = \pm(t \mp q_i)$ on the shell gives the matching; the weight along a matching is $\sum A_i A_{\sigma(i)} \le \sqrt{\sum A_i^2}\sqrt{\sum A_{\sigma(i)}^2} \le I$ per sign pair, and two sign pairs can land on the same $t$. (ii) Summing $A_i A_j$ over all ordered pairs and the four sign combinations counts $4S^2$, minus the excluded $t = 0$ diagonal contributions $2I$; Cauchy–Schwarz gives $S^2 \le nI$. □

**Lemma 4** ($\ell^2$ mass). $\displaystyle\sum_{t\neq 0} |w_t|^2 \;\le\; \big(\max_t |w_t|\big) \sum_{t\neq 0} |w_t| \;\le\; 2\lambda' I \cdot \lambda'(4S^2 - 2I) \;\le\; 8n\,(\lambda' I)^2.$

**Theorem 1** (Closed region of STEP-5B — derived). *Let $a(n,I) = \lambda'(4nI-2I)\,g(\hat r(I))$. For every admissible single-shell pattern with mode count $n$ and intensity $I$ such that $a(n,I) \le a_0 = 0.75$ and*

$$\delta(n,I) \;=\; \frac{2\lambda' I \cdot \lambda'(4nI-2I)\,J_{\max}}{2\,(1-a(n,I))} \;<\; 0.00432,$$

*the beyond-layer correction cannot overcome the weakest layer margin. At the production anchor the region contains all patterns with*

$$n \le n_{\max}(I): \qquad n_{\max}(10^{-4}) = 62, \quad n_{\max}(2{\times}10^{-4}) = 31, \quad n_{\max}(4{\times}10^{-4}) = 16,$$
$$n_{\max}(10^{-3}) = 6, \qquad n_{\max}(2{\times}10^{-3}) = 3.$$

*In particular every configuration in the v1.0 calibrated boxes is now DERIVED ($n = 12 \le 16$ at $I = 4{\times}10^{-4}$, margin ratio $\ge 13$).*

*Proof.* Combine Lemmas A–D: $\|X\| \le a(n,I)$ by Lemma A with the Lemma-C′ $\ell^1$ cap; Lemma B with the Lemma-D $\ell^2$ mass gives $|\delta F_{\text{off}}| \le \delta(n,I)$; the displayed inequality is then checked pointwise on the integer grid up to the stated $n_{\max}$ (script v1.1.1, 54/54 asserts; monotonicity of $n_{\max}$ in $I$ asserted). □

**Lemma 5** (Sphere additive energy — diagonal/off-diagonal split). *Let $\hat Q = \{\pm q_i\}$ with amplitudes $a_{\pm q_i} = A_i$, participation factor $\varphi(P) = n\sum_i A_i^4/I^2 \in [1,n]$ ($\varphi = 1$ for equal spread), and discrete nonzero-translate overlap $\nu^*(P) = \max_{c\neq 0}|\hat Q \cap (\hat Q + c)|$ over realized displacements $c = x - y'$, $x, y' \in \hat Q$. Then*

$$\sum_{t\neq 0} |w_t|^2 \;\le\; 4\,\lambda'^2 I^2\big(\varphi(P) + \nu^*(P)\big).$$

*Proof.* Write $w_t/\lambda' = (f{*}f)(t)$ with $f = \sum_{u\in\hat Q} a_u\delta_u$. Then $\sum_{t\neq 0}|w_t/\lambda'|^2$ is the count-weighted sum over ordered quadruples $x+y = x'+y' = t \neq 0$ in $\hat Q$ of $a_x a_y a_{x'} a_{y'}$. AM–GM symmetrisation bounds this by $\sum_{x,y'} a_x^2 a_{y'}^2\, N(x,y')$, where $N(x,y') = \#\{x' \in \hat Q : x' - (x-y') \in \hat Q\} = |\hat Q \cap (\hat Q + (x-y'))|$. The diagonal pairs $x = y'$ give $N = |\hat Q| = 2n$ against $\sum_x a_x^4 = 2\sum_i A_i^4$, i.e. $4\lambda'^2 I^2 \varphi$. The off-diagonal pairs have $N \le \nu^*$ against $\sum_{x\neq y'} a_x^2 a_{y'}^2 \le (2I)^2$. □

**Corollary 1** (Transversal $n$-free bound). *For every admissible single-shell pattern with $\nu^* \le 4$ and $\varphi \le 1+\epsilon$ (in particular generic/transversal configurations: measured $\nu^* = 2$ on random shells $n = 12, 24$), the beyond-layer envelope is $n$-FREE: at the anchor, with $c_E = 4(\varphi + \nu^*) \le 20$,*

$$|\delta F_{\text{off}}| \le \frac{c_E\,(\lambda' I)^2 J_{\max}}{2\,(1-a_E)} \;<\; 0.00432,$$

*with margin ratios $131\times/16\times/2\times$ at $I = 4{\times}10^{-4}/10^{-3}/2{\times}10^{-3}$, where the spectral control $a_E$ uses the E-route operating envelope (at most $\nu^* + 2$ heavy transfers per row, each of weight $\le 2\lambda' I$) — the row-count statement is the remaining G1″ item for this class.*

**Proposition 1** (Ring family — exact closed form). *For the canonical degenerate family (equal-amplitude two-ring pattern: a regular $n$-gon at height $z = q_0\cos\theta$ together with its antipodal image, $A_i^2 = I/n$, any $\theta \in (0,\pi/2)$),*

$$\sum_{t\neq 0}|w_t|^2 = c_{\text{ring}}(n)\,(\lambda' I)^2, \qquad c_{\text{ring}}(n) = \begin{cases} 14 - 18/n, & n \text{ even},\\ 8 - 6/n, & n \text{ odd},\end{cases}$$

*so $c_{\text{ring}}(n) < 14$ for every $n$. In particular the saturation observed numerically ($11.75 \to 13.72$ over $n = 8..64$) is exact.*

*Proof.* Write $e_k = (\cos\theta_k, \sin\theta_k)$, $\theta_k = 2\pi k/n$. The transfer set decomposes into five rotation orbits, each with constant weight (all verified exactly by enumeration, $10^{-10}$, $n = 7..64$):

- O1 (*$n$ even only*) heavy axial $t = (0, 0, \pm 2z)$: every same-ring pair $(k, k+n/2)$ has $e_k + e_{k+n/2} = 0$, so ALL $n$ ordered pairs collapse onto one transfer per ring: $w = \lambda' n A^2 = \lambda' I$; two transfers contribute $2(\lambda' I)^2$.
- O2 same-ring generic chord ($d \neq 0, n/2$): only the two orders of one pair reach a given $t$, $w = 2\lambda' A^2$; $2n(n/2-1)$ transfers (even) contribute $(4 - 8/n)(\lambda' I)^2$.
- O3 same-ring $d = 0$ ($t = \pm 2u_k$): $w = \lambda' A^2$, $2n$ transfers, $(2/n)(\lambda' I)^2$.
- O4 cross-ring generic ($t_z = 0$): for $n$ even the antipodal index shift $e_{k+n/2} = -e_k$ doubles the representation count ($u_i - u_j = u_{j+n/2} - u_{i+n/2}$), so $w = 4\lambda' A^2$ on $n(n/2-1)$ transfers: $(8 - 16/n)(\lambda' I)^2$; for $n$ odd there is no shift, $w = 2\lambda' A^2$ on $n(n-1)$ transfers: $(4 - 4/n)(\lambda' I)^2$.
- O5 (*$n$ even only*) cross-ring $d = n/2$ ($t = 2\rho e_k$, self-paired): $w = 2\lambda' A^2$, $n$ transfers, $(4/n)(\lambda' I)^2$.

Summing: even $2 + (4 - 8/n) + 2/n + (8 - 16/n) + 4/n = 14 - 18/n$; odd $(4 - 8/n + 4/n)_{\text{O2,O3 analogues}} + (4 - 4/n) = 8 - 6/n$. Each collapse statement is elementary regular-polygon geometry (uniqueness of chord representations); the orbit combinatorics is independent of $\theta$ because the heights only label the orbits and never merge them ($z \neq 0$). $\square$

**Lemma 6** (Collar heavy-mass bound — provable constant). *Let $P$ be $\kappa$-balanced ($\max_i A_i \leq \kappa\sqrt{I/n}$). For any row momentum $k$ and collar width $\delta \in (0, 2.1]$, with $S_\delta := \{p : \big||p| - q_0\big| < \delta q_0\}$ and $\nu_S^{\neq}(\delta) := \max_{c\neq 0} |\hat{Q} \cap (c + S_\delta)|$,*

$$\sum_{t\in T:\, k+t\in S_\delta} w_t \;\leq\; 2\lambda' I\big(\kappa + \nu_S^{\neq}(\delta)\big).$$

*Proof.* Each $w_t = \lambda' \sum_{(u,v):\, u+v=t} A_u A_v$ over ordered matched pairs; summing over heavy $t$ gives the pair sum over $E = \{(u, v) : u + v \in (-k + S_\delta)\}$. The centre family $c_u = -k - u$ isolates the single possible $c_u = 0$ term ($u = -k \in \hat{Q}$), whose mass is $A_{-k} \sum_v A_v \leq \kappa\sqrt{I/n}\sqrt{4nI} = 2\kappa I$ — the on-pattern rows $k = -u$ genuinely see all $2n$ partners (every $v \in \hat{Q}$ is on-shell), but their *mass* is $n$-free. For $c_u \neq 0$ both bipartite degrees obey $\deg(u) = |\hat{Q} \cap (c_u + S_\delta)| \leq \nu_S^{\neq}(\delta)$ and $\deg(v) \leq \nu_S^{\neq}(\delta)$, so the bilinear Cauchy–Schwarz bound gives $\sqrt{\nu 2I}\sqrt{\nu 2I}/\ldots \leq 2I\nu_S^{\neq}(\delta)$. Multiply by $\lambda'$. $\square$

**Registered negative result (row route).** Applying Lemma F over the collar ladder $\delta \in \{0.05, 0.1, 0.2, 0.4, 0.8, 2.1\}$ (the last rung exhausts all transfers, $|t| \leq 2q_0$, so no $\ell^1$ remainder survives) yields the row certificate $a_{\text{prov}}(P) = \sum_j 2\lambda' I(1 + \nu_j)/\sqrt{\hat{r} D_j}$. At the production anchor $I = 4 \times 10^{-4}$ this gives $a_{\text{prov}} = 2.20$ ($n = 12$) and $4.68$ ($n = 24$): *the Gershgorin/collar-ladder row route cannot certify $a < 1$ in the thin-spread regime with any constant this proof can currently supply.* The exploratory $\sqrt{\nu}$-constant scan (which would certify $a < 1$ up to $n \approx 30$) is NOT a theorem — verify-loop catch #5, caught by the devil's-advocate pass before registration. The scan values are retained in the artefact only as the bound *shape* the moment route must reproduce.

**Designated attack G1″-M4 (fourth moment).** Since $\|X\|^{2m} \leq \operatorname{tr} X^{2m}$ per volume, and $m = 1$ is Lemma E, the next rung is $m = 2$: $\operatorname{tr} X^4 = \sum w_{t_1} w_{t_2} w_{t_3} w_{t_4} \mathcal{J}_4$ over closed transfer quadruples $t_1 + t_2 + t_3 + t_4 = 0$, controlled by the four-fold additive energy $E_4(\hat{Q}) = \#\{\text{realized closed quadruples}\}$. Transversality ($\nu^* \leq 4$) caps $E_4 = O(n^2 \operatorname{poly}(\nu^*))$ against the generic $(2n)^3$, which is exactly the $n$-free gain the row route lacks. This is a multi-turn theorem and is registered as the single remaining mathematical gap of G1″(row).

**Lemma 7** (Sextic-vertex envelope ($\varepsilon_4$)). *On the closed region of the theorem above ($n \le n_{\max}(I)$, hence $nI \le 6.2 \times 10^{-3}$), the sextic vertex $6v$ contributes to any matched transfer weight at most a multiplicative correction $\varepsilon_4 = 60\, v\, nI/\lambda' \le 0.16$ relative to the quartic channel, so all sextic-class transfers fold into the Lemma-B envelope as a factor $(1+\varepsilon_4)$.*

**Lemma 8** ($\sigma$-channel completeness). *The $\sigma(x)$-inhomogeneity vertex (Math429 geometry) enters the off-diagonal operator $W$ only through the matched second-cumulant combination already counted in $\lambda' = 3u + 30vM$; the identity is exact (machine-verified to $10^{-12}$ against the Bloch coupling spectrum). No additional transfer class exists at second-cumulant order.*

**Lemma 9** (Two-shell suppression). *Transfers coupling the $q_0$ shell to the second BCC shell $q_1$ see the denominator floor $D \ge \hat{r} + C(q_1^2 - q_0^2)^2 = 0.5186 = 1.70\,\hat{r}$ at the anchor, so each cross-shell leg carries an extra factor $\le 1/\sqrt{1.70}$ inside Lemma A/B envelopes; the certified margins absorb this with the stated factor.*

**Theorem 2** (Composite glue at the $\ell^2$ level). *Let $P = \bigsqcup_\alpha P_\alpha$ be a cluster decomposition with pairwise cross-overlap $\nu_{\text{cross}} := \max_{\alpha\neq\beta} \max_{c\neq 0} |\hat{Q}_\alpha \cap (\hat{Q}_\beta + c)| \le 4$ (distinct circles meet any translate sphere pair in at most 4 points). Then*

$$\operatorname{tr} X^2(P) \;\le\; \sum_\alpha \operatorname{tr} X^2(P_\alpha) \;+\; \sum_{\alpha\neq\beta} 4\lambda'^2 I_\alpha I_\beta \big(\varphi_{\alpha\beta} + \nu_{\text{cross}}\big) J_{\max},$$

*with the within-cluster terms supplied exactly (ring closed forms) or by Lemma E. The certificate is validated on a ring–ring–random composite: measured $c = 9.27$ against certificate $34.23$.*

**Theorem 3** ($P^2$-representation and structural spectral floor). *Let $P := \sum_{u\in\hat{Q}} A_u S_u$, where $S_u$ is the shift $(S_u f)(k) = f(k+u)$ and $A_{-u} = A_u > 0$. Then $P = P^\dagger$ and the off-diagonal transfer operator of the second-cumulant Bloch Hessian obeys the exact operator identity*

$$W \;=\; \lambda'\big(P^2 - 2I\,\mathrm{Id}\big),$$

*i.e. the matched transfer weights $w_t = \lambda' \sum_{u+v=t} A_u A_v$ are precisely the off-diagonal coefficients of $\lambda' P^2$, and the $t = 0$ coefficient $2\lambda' I$ is precisely the diagonal Hartree dressing $\hat{r} - r_R$. Consequently, with $D_0(k) := r_R + C(k^2 - q_0^2)^2 > 0$:*

$$D + W \;=\; D_0 + \lambda' P^2 \;\ge\; D_0 \;>\; 0 \qquad \textit{(unconditional, every admissible pattern, every } n\textit{)},$$

*and the spectrum of $X = D^{-1/2} W D^{-1/2}$ is bounded below by*

$$X \;\ge\; -a_0, \qquad a_0 \;:=\; \sup_k \frac{2\lambda' I}{D(k)} \;=\; \frac{2\lambda' I}{\hat{r}},$$

*which is $n$-free and pattern-free ($a_0 = 0.0207$ at $I = 4\times 10^{-4}$; $a_0 = 0.0502$ at $I = 10^{-3}$).*

*Proof.* $P^\dagger = \sum_u A_u S_{-u} = P$ by the symmetry of $\hat{Q}$ and $A_{-u} = A_u$. Expanding $P^2 = \sum_{u,v} A_u A_v S_{u+v}$ collects, for each $t \neq 0$, exactly the ordered matched pairs defining $w_t/\lambda'$, and at $t = 0$ the coefficient $\sum_u A_u^2 = 2I$ — the identity is coefficient-wise exact (machine cross-check: max mismatch 0 over all realized transfers; the $\ell^1$ identity $\lambda'(4S^2 - 2I)$ of Lemma C′ is this structure in $\ell^1$ disguise). For the floor, for any $f$ with $g := D^{-1/2} f$:

$$\frac{\langle f, (1{+}X) f\rangle}{\|f\|^2} = \frac{\langle g, (D_0 + \lambda' P^2)\, g\rangle}{\langle g, Dg\rangle} \;\ge\; \inf_k \frac{D_0(k)}{D(k)} = 1 - \sup_k \frac{2\lambda' I}{D(k)} = 1 - \frac{2\lambda' I}{\hat{r}},$$

using $\lambda' > 0$ and $P^2 \ge 0$. The supremum is attained on shell. $\square$

**Corollary 2** (Envelope with the structural floor). *For every admissible pattern,* $\log(1+x) \geq x - x^2/(2(1-a_0))$ *holds on the entire spectrum of* $X$*, hence*

$$\left|\delta F_{\mathrm{off}}(P)\right| \;\leq\; \frac{\operatorname{tr} X^2/V}{4\,(1-a_0)} = \frac{\sum_t w_t^2\, J(|t|)}{4\,(1-a_0)}.$$

*The Gershgorin row condition* $a(P) < 1$ *is no longer needed anywhere in the chain; gap G1''-M4 (spectral control in the thin-spread regime) is* ***CLOSED BY STRUCTURE****. The v1.6 registered negative result (the row/collar-ladder route fails with its provable constant) stands as the route history that motivated this theorem.*

**Corollary 3** (Enlarged closed region). *Combining the corollary with the general* $\ell^2$ *mass bound (Lemma D,* $\sum_t w_t^2 \leq 8n(\lambda' I)^2$*) gives STEP-5B on* $n \leq N_{\max}(I)$ *with*

$$N_{\max}(I) = \left\lfloor \frac{4(1-a_0)\,m}{8(\lambda' I)^2 J_{\max}} \right\rfloor = 12133/3017/746/115/27 \quad \textit{at } I = 10^{-4}/2{\times}10^{-4}/4{\times}10^{-4}/10^{-3}/2{\times}10^{-3},$$

*(*$m = 0.00432$ *the weakest interval margin), versus* 62/31/16/6/3 *for the superseded Gershgorin route — a* $\times 46$ *enlargement at the production anchor. Transversal patterns remain covered n-free (Lemma E corollary, now with* $(1-a_0)$*); the canonical ring family remains covered exactly.*

*Remark* 1 (Robustness under G2 corrections). The sextic class enters multiplicatively within the certified region (Lemma H, $\varepsilon_4 \leq 0.16$), shifting the floor at most to $a_0(1+\varepsilon_4) \leq 0.025$ at the anchor; the $\sigma$-channel adds no new transfer class (Lemma I); two-shell legs are suppressed (Lemma J). The floor statement is at the same second-cumulant bookkeeping scope as the whole note (the Nambu/anomalous block is inside the Math427–434 five-reading bookkeeping that calibrates $X$; its independent re-verification is listed in section 6).

**Lemma 10** (Position-space representation). *In position space* $P$ *is multiplication by the condensate profile* $F(x) = \phi_n(x) = \sum_{u\in\hat{Q}} A_u e^{iux}$ *(real), so the* $P^2$*-representation reads*

$$W \;=\; \lambda'\bigl(F(x)^2 - 2I\bigr), \qquad D + W \;=\; D_0 \;+\; \lambda' F(x)^2,$$

*a* multiplication operator *added to* $D_0$*: the floor* $D{+}W \geq D_0$ *holds pointwise, which is the elementary form of the structural theorem. This also* ***discharges the anomalous-block objection*** *(δ of section 6): the TECT order parameter is a real scalar field; its second-cumulant fluctuation operator is the single real symmetric Hessian* $\hat{K}[\phi_n] = \hat{r}_{\mathrm{eff}} + C(q^2-q_0^2)^2 + (3u+30vM)\phi_n^2 + \ldots$ *(Math427 structure) — there is no independent Nambu/pairing block at this scope; the* $\pm k$ *Bloch pairing is already encoded in the symmetric* $\hat{Q}$ *with* $A_{-u} = A_u$*.*

**Proposition 2** (Parseval reformulation of G1'''-AE). $\sum_{t\neq 0} w_t^2 \;=\; \lambda'^2\bigl(\langle F^4\rangle - 4I^2\bigr)$*, where* $\langle\cdot\rangle$ *is the spatial average. Hence the remaining gap G1'''-AE is exactly the* discrete sphere $L^4$-extension inequality $\langle F^4\rangle \leq K\,I^2$ *for sphere-supported trigonometric sums — the Stein–Tomas exponent* $q = 2(d+1)/(d-1) = 4$ *at* $d = 3$*. The mechanism that keeps* $K$ *finite on all certified families is sphere curvature: a sum* $u+v=s$ *confines the pair to the circle* $C_s = S\cap(s-S)$*, and a fixed sum on a circle pins the unordered pair (fiber rigidity), up to the axial exceptions quantified below. (Machine check: the identity holds to MC accuracy* 0.5%/7.5% *on random/ring test patterns.)*

**Theorem 4** (Universal single-circle bound — sharp constant 14). *Let all pattern points lie on ONE circle* $C\subset S$ *(any height, any n, arbitrary positive amplitudes,* $I_c = \sum_i A_i^2$*). Then*

$$\sum_{t\neq 0} w_t^2 \;\leq\; 14\,\lambda'^2 I_c^2,$$

*and the constant is sharp: the equal-amplitude ring family attains* $14 - 18/n \to 14$*.*

*Proof.* Write $\hat{C} = C \cup (-C)$ (top/bottom copies). Sum fibers $\{(\hat{u}, \hat{v}) : \hat{u} + \hat{v} = t\}$ classify as: (i) top–top or bottom–bottom with $t$ non-axial: a fixed sum on a circle determines the unordered pair (two ordered pairs) — by $(\sum_{\le 2} a)^2 \le 2 \sum a^2$, their total is $\le 2(\sum_{\text{top}} A^2)^2 + 2(\sum_{\text{bot}} A^2)^2 = 4I_c^2$; (ii) top–bottom and bottom–top combined: difference fibers on a circle carry $\le 4$ ordered pairs (the $(d, m) \sim (-d, m+n)$ index identification plus ordering), total $\le 4 \cdot 2I_c^2 = 8I_c^2$; (iii) the two axial transfers $t = \pm 2z\hat{z}$ (present for antipodally-paired angle sets, e.g. even equal-angle rings): $\Phi_\pm \le I_c$ each by Cauchy–Schwarz, total $\le 2I_c^2$. Summing: $4 + 8 + 2 = 14$. Sharpness from the exact ring closed form $c_{\text{ring}}(n) = 14 - 18/n$ (five-orbit proposition above), whose orbit masses saturate (i)–(iii) asymptotically. Machine verification: 8 random-amplitude/height/$n$ circles give $K \le 12.23$; the $n = 64$ equal-amplitude ring gives 13.7187. □

*Remark* 2 (Consequences and the remaining hard class). (a) The equal-amplitude caveat of the ring-family proposition is REMOVED: single-circle patterns are closed universally, so the residual corner excludes all of them. (b) The suspected-hard class for the absolute-$K$ conjecture — coaxial circle pairs, whose sumset concentrates on an annulus (many realized circles through point pairs) — was probed as a pre-registered falsification test: $K = 9.9/10.4/10.7$ at $2{\times}8/2{\times}16/2{\times}32$, bounded (supporting evidence, NOT a proof). (c) The corner therefore consists of high-$n$, multi-circle, non-transversal patterns without a $\nu_{\text{cross}} \le 4$ decomposition — exactly the configurations where the discrete $L^4$ problem is open.

**Theorem 5** (Antipodal-carrier partition)**.** *For every ordered pair $(\hat{u}, \hat{v}) \in \hat{Q}^2$ with $s = \hat{u} + \hat{v} \ne 0$ there is exactly ONE circle on the sphere having $\{\hat{u}, \hat{v}\}$ as an antipodal pair: the* carrier *$C_s = S \cap (s - S)$, centred at $s/2$ (for $\hat{v} = \hat{u}$ the carrier degenerates to the point $\hat{u}$). Distinct $s$ give distinct carriers, so the set of ordered pairs PARTITIONS by carriers, and*

$$\Phi_s \;=\; \Psi_{C_s} := \sum_{\substack{\hat{u} \in \hat{Q} \cap C_s \\ s - \hat{u} \in \hat{Q}}} A_{\hat{u}} A_{s - \hat{u}}, \qquad \sum_{t \ne 0} w_t^2 \;=\; \lambda'^2 \sum_C \Psi_C^2.$$

*(Machine check: $\ell^1$ and $\ell^2$ reconstructions from the carrier partition match the transfer sums exactly, $10^{-12}/10^{-15}$.)*

**Theorem 6** ($\nu^* = \mu_C$: transversality IS circle-thinness)**.** *The Lemma-E translate-overlap parameter $\nu^* = \max_{c \ne 0} |\hat{Q} \cap (\hat{Q} + c)|$ equals $\mu_C := \max_{\text{circles } C \subset S} |\hat{Q} \cap C|$, the maximal number of pattern points on a single circle. (A shifted-shell overlap $S \cap (c + S)$ is the circle centred $c/2$; conversely every circle on $S$ arises this way.) Hence the transversal route (Lemma E) and the circle-decomposition route are controlled by ONE parameter. Verified $\nu^* = \mu_C$ on random (2), ring ($16 = n$), coaxial (12) test patterns.*

**Theorem 7** (Coaxial-class closure)**.** *Let $P$ be supported on a union of coaxial circles $C_1, \dots, C_m$ (axis $\hat{z}$, arbitrary heights, radii, point sets, amplitudes; pattern points pairwise distinct). Then every off-axis carrier holds at most 2 unordered (4 ordered) antipodal pattern pairs, and*

$$\sum_{t \ne 0} w_t^2 \;\le\; \lambda'^2 \Big( 14 \sum_j I_j^2 + 16\, I^2 \Big) \;\le\; 30\, \lambda'^2 I^2.$$

*Proof.* A carrier with centre $c$ pairs $\hat{u}$ with $2c - \hat{u}$, i.e. pairs points of $C_j$ with points of $C_k \cap (2c - C_j)$. The reflected circle $2c - C_j$ is a circle of the same radius with centre $2c - c_j$ and parallel plane; if $2c - C_j \ne C_k$ the two circles meet in at most 2 points, giving at most 2 unordered pairs per cluster pair $(j, k)$; for coaxial clusters at most one cluster pair can realize any fixed off-axis $c$ (heights are pinned: $z_c = (z_j + z_k)/2$ and the in-plane offset forces $j \neq k$ uniquely for distinct height pairs), so off-axis carriers hold $\leq 4$ ordered pairs. The coincidence $2c - C_j = C_k$ forces $c_k = 2c - c_j$ with parallel normals and equal radii, hence $c$ ON the axis: such carriers are on-axis. On-axis carriers with $c = z\hat{z}$ pair each circle with its equal-radius mirror partner at height $2z - z_j$; the within-circle axial case ($C_k = C_j$) is inside the universal single-circle theorem ($14I_j^2$); the distinct-circle mirror case at $\pm z$ equal radius has $c = 0$, i.e. $s = 0$, excluded from all $t \neq 0$ sums. Collecting: within-circle $\leq 14 \sum_j I_j^2$; off-axis carriers, by the $(\sum_{\leq 4} a)^2 \leq 4 \sum a^2$ fiber bound and the pair partition, contribute $\leq 4 \cdot 2 \cdot \sum_{\text{cross pairs}} (A_{\hat{u}} A_{\hat{v}})^2 \leq 8I^2$; the diagonal (pair-with-itself and reversal) terms contribute $\leq 8I^2$ as in the Parseval proposition. Total $\leq 14 \sum I_j^2 + 16I^2 \leq 30I^2$. Measured: $K = 9.40$ (three distinct radii), $K = 10.73$ (equal-radius $\pm z$ mirror configuration, nondegenerate). The pre-registered suspected-hard class is CLOSED. □

*Remark* 3 (Honest falsification record: H-GEN(2) is false; the sharpened residual). The naive extension hypothesis — "for arbitrary multi-circle unions every non-cluster carrier holds $\leq 2$ unordered pairs" — is FALSE: a symmetric $\text{ring}_8 \cup \text{ring}_{10} \cup \text{random}_6$ union exhibits a non-cluster carrier with 10 ordered pairs (ring difference carriers and cross-cluster coincidences). The probe was pre-registered and did its job. Consequently the general multi-circle bound must carry the *carrier-richness parameter*

$$p_0(P) := \max_{C \notin \{\text{cluster circles}\}} \#\{\text{ordered antipodal pattern pairs on } C\},$$

$$\sum_{t \neq 0} w_t^2 \leq \lambda'^2 \Big( 14 \sum_j I_j^2 + 8I^2 + 2p_0 I^2 \Big),$$

and gap G1‴-AE is SHARPENED to: bound $p_0$ (or its mass-weighted form) class-wide. All probed configurations keep $K < 32$ regardless ($K_{\text{mixed}} = 9.08$). Verify-loop catch #6: the first mirror test configuration was degenerate (duplicate points from $\text{ring}(\pi - 0.7) = -\text{ring}(0.7)$ point-for-point at $n = 10$) — a test artifact, rebuilt with a phase offset and a nondegeneracy assert.

**Lemma 11** (Off-axis carrier multiplicity — repaired with $H^*$)**.** *For a coaxial union, an off-axis carrier centre $c$ can receive antipodal pairs from a cluster pair $(j,k)$ only if (i) $z_j + z_k = 2z_c$, (ii) the reflected circle $2c - C_j$ meets $C_k$ (annulus condition on $|c_{xy}|$), and (iii) the $\leq 2$ intersection points are realized pattern points. With $H^*(c)$ the number of cluster pairs satisfying (i)–(iii),*

$$\Psi_{C_c} \;\leq \sum_{(j,k) \in H^*(c)} \psi_{jk}(c), \qquad \psi_{jk}(c) \leq 2\,(\textit{max pair mass on } C_j \times C_k),$$

*and the coaxial constant carries $H^* := \max_c H^*(c)$ explicitly. The v1.9 proof implicitly used $H^* = 1$; the AUDIT (operator verdict #8) with FORCED height-sum coincidences (arithmetic-progression heights, $m = 3/5/7$ ring stacks) measures off-axis carriers at 4 ordered pairs throughout — i.e. $H^* = 1$ persists because the in-plane condition (ii)–(iii) separates cluster pairs that share a height sum; K DECREASES with $m$ (9.25/8.75/8.54, within-circle masses diluting as $\sum_j I_j^2 = I^2/m$), and the random-amplitude stack gives $K = 8.31$. The theorem statement is now hypothesis-explicit: $K_{\text{coax}} \leq 14 \sum_j I_j^2 / I^2 + 8 + 8H^*$, with $H^* = 1$ measured on all probes and $H^* \geq 2$ requiring a four-circle reflection coincidence that the audit could not produce.*

**Theorem 8** (Rectangle reformulation)**.** *Two distinct antipodal pairs of the same carrier are two diameters of one circle, i.e. a RECTANGLE inscribed in a circle on the sphere. Hence*

$$\sum_{t \neq 0} w_t^2 = \lambda'^2 \Big( \underbrace{\textstyle\sum_{\text{pairs}} \textit{(diagonal terms)}}_{\leq 8I^2 \ \textit{unconditional}} + \underbrace{\textstyle\sum_{\textit{rectangles } R \subset \hat{Q}} A_{\hat{u}} A_{\hat{v}} A_{\hat{u}'} A_{\hat{v}'}}_{\textit{weighted rectangle count}} \Big),$$

*(self-pair carriers $\hat{u} = \hat{v}$ enter the diagonal with single multiplicity). G1'''-AE is exactly the weighted rectangle-count bound. Machine check: the split reconstructs $\sum w_t^2$ to $10^{-15}$; rand$_{12}$ carries $-0.17I^2$ rectangle mass (self-pair convention), ring$_{16}$ carries $+5.13I^2$.*

**Theorem 9** (Triple count and the unconditional $n^{5/2}$ bound)**.** *Two distinct circles meet in at most two points, so three distinct points lie on at most one circle: $\sum_C k_C(k_C-1)(k_C-2) \le 2n(2n-1)(2n-2)$, i.e. $\sum_C k_C^3 = O(n^3)$. Combined with the carrier-partition constraints $\sum_C p_C \le 2n(2n-1)$ and $p_C \le k_C$, dyadic optimization yields the unconditional ordered-rectangle bound*

$$R(\hat{Q}) \;=\; \sum_C p_C(p_C-1) \;=\; O\bigl(n^{5/2}\bigr),$$

*(single rich circles contribute only $O(n^2)$; the extremal profile is $\sim n/\varepsilon$ circles of richness $\varepsilon n$ with $\varepsilon \le n^{-1/2}$ forced by the triple count).*

**Corollary 4** ($\sqrt{n}$ energy bound and the upgraded region)**.** *For $\kappa$-balanced patterns, $\sum_{t\neq 0} w_t^2 \le \lambda'^2 I^2(8 + c_R\kappa^4\sqrt{n})$ with $c_R$ the rectangle prefactor (measured $c_R \approx 4$ on adversarial configurations; first-principles derivation registered as an open sub-task). The $K$-budget at the anchor ($4(1-a_0)m/((\lambda' I)^2 J_{\max}) = 5972$ at $I = 4\times 10^{-4}$) then closes all $\kappa$-balanced patterns with*

$$n \;\lesssim\; 2.2\times 10^6 \;/\; 5.3\times 10^4 \;/\; 2.8\times 10^3 \qquad \text{at } I = 4{\times}10^{-4}/10^{-3}/2{\times}10^{-3},$$

*superseding $N_{\max}(I) = 746/115/27$ by three orders of magnitude.*

## 4. The reduction (precise form of STEP-5B)

**Proposition 3** (Pattern-generic reduction — structural form)**.** *For every admissible pattern $P$,*

$$\bigl|\delta F_{\text{off}}(P)\bigr| \;\le\; \frac{\sum_t w_t^2\, J(|t|)}{4\,\bigl(1-a_0\bigr)}, \qquad a_0 = \frac{2\lambda' I}{\hat{r}} \quad (\textit{structural floor; } a_0 = 0.0207 \textit{ at } I = 4{\times}10^{-4}),$$

*with NO pattern-dependent spectral condition: the floor is unconditional. Consequently STEP-5B holds wherever $\sum_t w_t^2 J(|t|) < 4(1-a_0)\,m$ ($m$ the applicable margin), and the full gate reduces to the single class-wide $\ell^2$-mass question G1'''-AE of the Parseval proposition.*

*Remark* 4 (Superseded auxiliary route). The original Gershgorin reduction (rows $a(P) := W_{\ell^1}(P)\,g(\hat{r}) < 1$ and the envelope with $(1-a(P))$) remains valid where it applies but is SUPERSEDED by the structural form above: the v1.6 registered negative result showed its provable row constants cannot reach the thin-spread regime, and the structural floor makes the condition unnecessary. It is retained in section 3 as route history and as an independent cross-check on the certified boxes.

## 5. Numerical verification

Script `codes/vacuum/beyond_layer_gershgorin_bound.py` (v1.8.0; 155/155 self-test asserts PASS; artefact `claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/result.json`).

| Quantity | Value | Check |
|---|---|---|
| anchor $r_R$, $M_R$, $\lambda'$, $64Cq_0^4$ | 0.3045257, 0.109414, 8.0551, 13.698 | match locked constants |
| Math434 calibration ($A = 0.01$): $\hat{r}'$, row terms | 0.30613; $2.624{\times}10^{-3} + 3.879{\times}10^{-4}$ | reproduces $2.633{\times}10^{-3} + 3.893{\times}10^{-4}$; $\|X\| \le 3.1{\times}10^{-3}$ |
| $J(\|t\|)$, $\|t\|/q_0 \in \{0.5, 1, \sqrt{2}, \sqrt{3}, 2\}$ | $0.273, 0.218, 0.169, 0.133, 0.104$ | grid-refinement drift $< 6{\times}10^{-6}$; shell estimate 0.161 brackets $J_{\max}$ |
| calibrated box $n_{\mathrm{res}}$=12, $I = 4{\times}10^{-4}$ | $a = 0.143$, $\delta \le 2.38{\times}10^{-4}$ | margin ratio 18.2× |
| calibrated box $n_{\mathrm{res}}$=12, $I = 10^{-3}$ | $a = 0.347$, $\delta \le 1.95{\times}10^{-3}$ | margin ratio 2.2× |
| LAM second order | $\delta_{\mathrm{LAM}} = 6.7{\times}10^{-8}$ | margin/64,000 |
| matching lemmas on rings/random shells | exact $\ell^1$ identity; $m_t \le 2n$; $w_t \le 2\lambda' I$ | equality on rings to $10^{-12}$ |
| $n$-uniform $\ell^2$ evidence (rings 16–32) | $\sum w_t^2 = 12.9$–$13.5\,(\lambda' I)^2$ | vs $8n$ bound: 19× slack at $n$=32 |
| derived $n_{\max}(I)$ | 62/31/16/6/3 at $I = 10^{-4}..2{\times}10^{-3}$ | $a \le 0.75$; boxes of v1.0 now derived |
| Lemma E split (rings/random, $n \le 64$) | $c_{\mathrm{meas}} \le 4(\varphi + \nu^*)$ everywhere | random: $\nu^* = 2$, $c_{\mathrm{meas}} = 7.5$–$7.75 \le 12$ |
| ring family $c(n)$ | EXACT: $14 - 18/n$ (even), $8 - 6/n$ (odd) | closed form verified $10^{-10}$, $n = 7..64$ |
| transversal $n$-free corollary | ratios $131\times/16\times/2\times$ at $I = 4{\times}10^{-4}/10^{-3}/2{\times}10^{-3}$ | $\nu^* \le 4$, $\varphi \le 1$; $n$-FREE |
| row route, provable constant | $a_{\mathrm{prov}} = 2.20/4.68$ at $n = 12/24$ | $> 1$: NEGATIVE result, route refuted |
| G2 trio (Lemmas H/I/J) | $\varepsilon_4 \le 0.16$; $\sigma$ exact $10^{-12}$; floor ×1.70 | bookkeeping closed on region |
| composite glue | measured $c = 9.27$ vs certificate 34.23 | $\nu_{\mathrm{cross}} = 1 \le 4$ on test composite |
| $P^2$ identity | max mismatch 0 over all transfers | $t$=0 coefficient $= 2I$ exactly |
| spectral floor (sections) | min-eig $\ge -a_0$ on rings/composites/near-pairs | stressed section dim 75: $-0.0034 \ge -0.0207$ |
| $N_{\max}(I)$ structural | 12133/3017/746/115/27 | vs Gershgorin 62/31/16/6/3 |
| Parseval identity | MC dev 0.5% / 7.5% (rand12/ring16) | $\sum w^2 = \lambda'^2(\langle F^4\rangle - 4I^2)$ |
| single-circle universal | $K \le 12.23$ (8 random draws); 13.7187 at ring$_{64}$ | constant 14 SHARP |
| coaxial AE probe | $K = 9.9/10.4/10.7$ at $2{\times}8/16/32$ | bounded; pre-registered probe |
| carrier partition | $\ell^1/\ell^2$ reconstruction exact ($10^{-12}/10^{-15}$) | pair set partitions by carriers |
| $\nu^* = \mu_C$ | 2/16/12 on rand/ring/coaxial | one parameter, two routes |
| coaxial closure | $K = 9.40$ (distinct radii), 10.73 (mirror, nondegenerate) | off-axis carriers $\le 4$ ordered |
| H-GEN(2) probe | FALSIFIED: 10 ordered pairs on a non-cluster carrier | residual sharpened to $p_0$ |
| rectangle split | exact to $10^{-15}$; diag $\le 8I^2$ | ring$_{16}$: $+5.13I^2$ rectangles |
| AP-height audit | $H^* = 1$ at $m = 3/5/7$; $K = 9.25/8.75/8.54$ | forced coincidences separated in-plane |
| $\sqrt{n}$ corollary | $K$-budget 5972; $n \lesssim 2.2{\times}10^6$ at anchor | $\kappa$-balanced; $c_R$ measured |

Quantitative sanity check (mandatory): the LAM row terms agree with the Math434-audit figures to $\le 4{\times}10^{-6}$; $J$ obeys the analytic shell estimate $q_0/(8\pi\hat{r}^{3/2}\sqrt{C}) = 0.161$ within a factor 1.7; dimensions: $[w] = [\hat{r}]$, $[J] = [\hat{r}]^{-2}\,[k]^3$ consistent with $\delta F$ per unit volume.

The original v1.0 calibrated boxes are now DERIVED within the closed-region theorem. The original thin-spread residual has been sharpened to G1‴-AE + G-DEC (sub-route), with G1″-M4 closed by structure and G2 bookkeeping closed at second-cumulant level; the certified territory is $n \le N_{\max}(I) \cup$ transversal $\cup$ rings $\cup$ glued composites.

## 6. Devil's-advocate (three objections)

($\alpha$) *"$W_{\ell^1}$ grows without bound in $n$, so nothing class-wide is proven."* VALID-with-mitigation, now sharpened: the closed-region theorem converts the $\ell^1$ route into an explicit derived region (now $N_{\max}(I)$ via the structural floor); the residual has been sharpened through G1$'$ $\to$ G1$''$-M4 (closed by structure) $\to$ G1$'''$-AE on the high-$n$ multi-circle incidence corner, now reduced to the carrier-richness bound $p_0$. The class-wide statement remains open, hence T4 and not T5.

($\alpha'$) *"The author's own constants may be wrong."* Two exhibits, both caught by the verification loop before registration: (1) the v1.1.0 $\ell^1$ assert with the wrong constant $2S^2$ FAILED on every configuration; ring measurements matched the corrected identity $\lambda'(4S^2-2I)$ to $10^{-12}$. (2) the v1.2.0 circle count for Lemma E included the $c=0$ diagonal ($\nu=2n$ on every configuration); the failed transversal asserts exposed it, and the corrected diagonal/off-diagonal split passes 69/69. (3) catch #4: the first v1.4.x collar functional included the $c=0$ centre ($\nu=2n$ on every configuration); the on-pattern rows $k=-u$ explain the genuine $2n$ count, and the mass split of Lemma F resolves it. (4) catch #5: the exploratory $\sqrt{\nu}$ ladder constant was caught by this self-test BEFORE registration; the provable constant FAILS, and the failure is registered as a negative result rather than silently dropped.

($\delta$) *"The $P^2$ identity might hold for the code's $W$ but not for the physical Bloch Hessian (anomalous/Nambu block, pair-channel vertex)."* DISCHARGED in v1.8 (was VALID-with-mitigation in v1.7): the order parameter is a REAL scalar field; the second-cumulant fluctuation operator is the single real symmetric Hessian $\hat{K}[\phi_n]$ of Math427, and $W$ is the multiplication operator $\lambda'(\phi_n^2-\langle\phi_n^2\rangle)$ — there is no independent pairing block at this scope. The position-space lemma makes the floor pointwise and manifest.

($\varepsilon$) *"A finite-section check cannot verify an operator-norm statement."* DISMISSED with the correct logic direction: compressions can only RAISE the minimum eigenvalue, so sections cannot prove the floor — but they can FALSIFY it. Seven adversarial sections (rings at the axial resonance, composites, near-coincident pairs, a 75-dimensional stressed section) all sit ABOVE $-a_0$, with ring sections approaching it ($-0.0123$ vs $-0.0207$): the floor is consistent and near-sharp on the known worst family. The PROOF of the floor is the quadratic-form argument, not the sections.

($\zeta$) *"With G1$''$-M4 closed, why is the gate still open?"* Because the envelope still requires the $\ell^2$ mass $\sum w_t^2$, and its class-wide $n$-free bound (G1$'''$-AE) is exactly the corner left open; no closure is claimed.

($\eta$) *"The Parseval identity check is Monte Carlo with 7.5% deviation on rings — weak."* VALID-with-mitigation: the identity itself is two lines of algebra (Parseval applied to $F^2$); the MC is a smoke test against implementation error, slow to converge on rings because near-degenerate transfer directions equidistribute slowly in a finite box. The exact transfer-sum side is already verified to $10^{-12}$ against the orbit closed forms.

($\iota$) *"The $\sqrt{n}$ corollary's prefactor $c_R$ is measured, not derived."* VALID-with-mitigation: the EXPONENT $n^{5/2}$ is unconditional (triple-count optimization); only the constant is calibrated on adversarial configurations (rings, coaxial stacks, mixed unions, all $\leq 9.4$ total). The first-principles constant is registered as the companion sub-task of G1$'''$-AE; the closed-region figures quote the measured constant explicitly and are tagged $\kappa$-balanced.

($\beta$) *"$J(|t|)$ is numerical only."* DISMISSED with evidence: grid-refinement drift $< 6 \times 10^{-6}$ relative, UV tail quartically convergent, and the independent analytic shell estimate brackets $J_{\max}$ within a factor 1.7.

($\gamma$) *"The comparison uses the global minimum margin, but Proposition A's floors are interval-wise."* VALID-with-mitigation: using the weakest interval margin $+0.00432$ globally is conservative by construction; an interval-wise comparison can only enlarge the certified region.

## 7. Result footer

```
Result ID:                B5-BEYOND-LAYER-BOUND (serving gate STEP-5B)
Precise statement:        v1.9 stack PLUS: coaxial lemma repaired with
                          explicit H* (audit: forced height coincidences
                          give H* = 1; K decreases with m; weighted 8.31);
                          RECTANGLE REFORMULATION (off-diag energy =
                          weighted inscribed-rectangle count; diag <= 8I^2);
                          TRIPLE-COUNT THEOREM (sum k^3 = O(n^3)) =>
                          R = O(n^{5/2}) UNCONDITIONAL; kappa-balanced
                          K(n) <= 8 + c_R sqrt(n) => region upgraded to
                          ~2.2e6 / 5.3e4 / 2.8e3 modes. 155/155 asserts.
Scope:                    Second-cumulant bookkeeping scope (= Math434
                          calibration scope); anomalous-block objection
                          discharged (real scalar, single Hessian).
                          STEP-5B remains OPEN on the corner
                          {n > N_max(I), non-transversal, multi-circle
                          without nu_cross<=4 decomposition}: G1’’’-AE.
                          Equal-amplitude caveat on circles REMOVED.
Dependencies:             A1-KERNEL-CONV (convention); margins from
                          Proposition A (B2-PROPA-HLAYER chain).
Evidence grade:           ANALYTIC (theorem + lemmas) + EXECUTED
                          (identity/floor/region machine-verified) +
                          NEGATIVE (row route, stands as route history)
Reproduction command:     python codes/vacuum/beyond_layer_gershgorin_bound.py
Expected output:          claims 155/155 PASS; result.json under
                          claims/B5-BEYOND-LAYER-BOUND/runs/
                          260605-gershgorin-reduction/
Falsification gate:       Any admissible-pattern section with min-eig below
                          -a_0 (would refute the floor and the P^2
                          identity); any pattern in the certified territory
                          whose exact correction exceeds the envelope; any
                          pattern undercutting the +0.00432 band margin.
Tier before / after:      T4 / T4 (T5-candidacy flagged for operator:
                          the single remaining gap is G1’’’-AE on an
                          extreme corner)
No-overclaim statement:   This note does NOT close STEP-5B: the corner
                          (high-n, multi-circle, non-coaxial, rich
                          carriers) needs the class-wide bound on p0. No
                          tier action on B1/B2; Reading-H stays T5. The
```

anomalous/Nambu objection is DISCHARGED within the real-scalar second-cumulant Hessian scope - no independent pairing block is present at this scope; statements beyond that scope (higher cumulants) are not claimed.

Next required action: (i) first-principles rectangle prefactor c_R; (ii) the extreme-n corner n > ~1e6 at the anchor (rich-carrier non-coaxial patterns beyond the sqrt(n) region) - either the sharp R = O(n^2) conjecture or an admissibility-scale argument; then operator review for closure assessment.